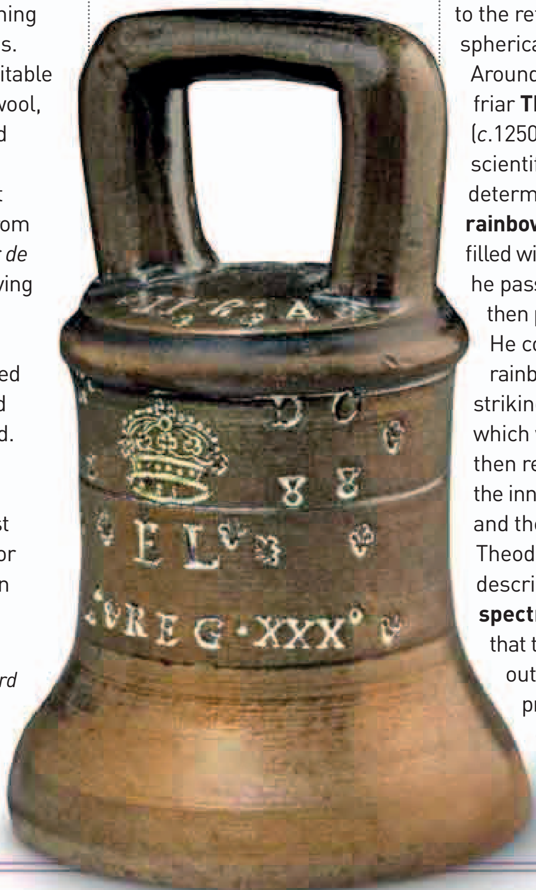
# 1300–10

The search for an explanation of the colors of the rainbow preoccupied medieval scholars, including Roger Bacon and Theodoric of Freiberg.

## 0.45 KILOGRAMS THE **MODERN** EQUIVALENT OF THE **AVOIRDUPOIS POUND**

**EARLY MEDIEVAL EUROPE USED A SYSTEM OF WEIGHTS** based on the Roman pound, which had 12 *unciae* (or ounces) and was mainly used for weighing pharmaceuticals and coins. A new set of measures suitable for bulky goods, such as wool, was introduced in England around 1303 (when it was mentioned in a charter). It was called **avoirdupois**, from the Norman French *Habur de Peyse* meaning "goods having weight" and was **based on a 16-ounce pound**—a measure that would be used for the next 700 years, and still is in parts of the world.

The avoirdupois pound probably originated in Florence, where an almost identical unit was in use for the weighing of wool. Soon supplementary weights were added, including the hundredweight (112 pounds), defined in an ordinance of 1309.



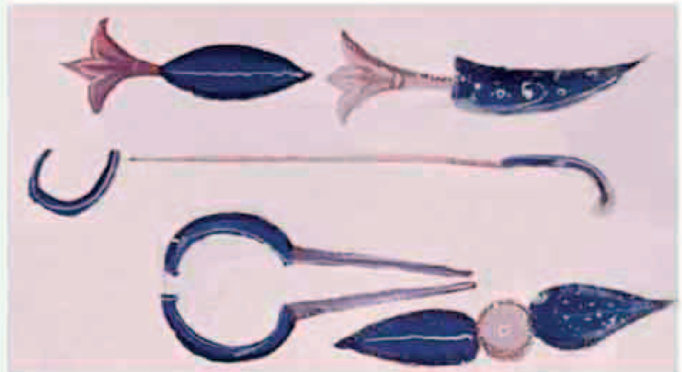
**Weighty issues**
*This is one of a set of standard avoirdupois weights that were issued by Elizabeth I of England in 1582. These weights were to remain the standard measure until the 1820s.*

The properties of the rainbow had fascinated philosophers from Aristotle to Roger Bacon, who thought that their color was due to the reflection of light from spherical raindrops in a cloud.

Around 1310, the Dominican friar **Theodoric of Freiberg** (*c*.1250–1311) carried out scientific experiments to determine the **origins of rainbows** by using glass balls filled with water through which he passed light, which was then projected onto a screen. He concluded that the rainbow was caused by light striking spherical raindrops, which was first refracted, then reflected internally on the inner surface of the drop, and then refracted again. Theodoric also properly described the **color spectrum**. He discovered that the light that projected out of his glass balls produced the same range of colors as a rainbow, and in the same order (red, yellow, green, and blue).

# 1311–16

The title page of Mondino de Luzzi's *Anatomia* shows a corpse on the dissecting table; it has already been opened up and had organs removed.



### EARLY SURGERY AND DISSECTION

Surgery took a long time to become established as a separate discipline, but written records since 1170 indicate a growing medical sophistication. By 1200, operations for bladder-stones, hernias, and fractures were routine. By the 14th century, surgeons were aware of the need to avoid infection, sometimes cleaning wounds with wine and closing them up as soon as possible.

**THE EARLIEST WRITTEN RECORDS OF WEIGHT-DRIVEN CLOCKS** feature in the Italian writer Dante Alighieri's book *Paradiso* (*Paradise*) (*c*.1313–21), although such clocks probably first appeared decades earlier. Weight-driven clocks use a weight to act as an energy storage device so that the clock can run for a certain period of time (such as a day or a week). Winding such clocks pulls on a cord that lifts the weight, which is affected by gravity and falls; the clock uses the potential energy as the weight falls to drive the clock's mechanism. The first clock faces were probably divided up according to the canonical hours (seven regulated times of prayer), which punctuated the church day. **Clocks showing 12 equal hours** were first recorded in 1330.

Among the first major surgical writers was **Henri de Mondeville** (*c*.1260–1316). A former military surgeon who came to teach medicine at Montpellier, de Mondeville had, by 1308, **begun to use anatomical charts and a model of a skull as aids to his teaching**. Around 1312, he

**1303 Avoirdupois system** of weights introduced in England

**1305–07** English physician John of Gaddesden describes the **pelican**, an instrument for extracting teeth

**1309 Hundredweight** is added to avoirdupois system

**c.1310** Theodoric of Freiberg carries out experiments on the **color spectrum** and rainbow

**1312** French surgeon **Henri de Mondeville** publishes a major manual on surgery entitled *Surgery*

**1313–21** Italian writer Dante Alighieri mentions a **weight-driven clock** in his book *Paradise*